TBWB

Angular Fundamentals Module - Databinding

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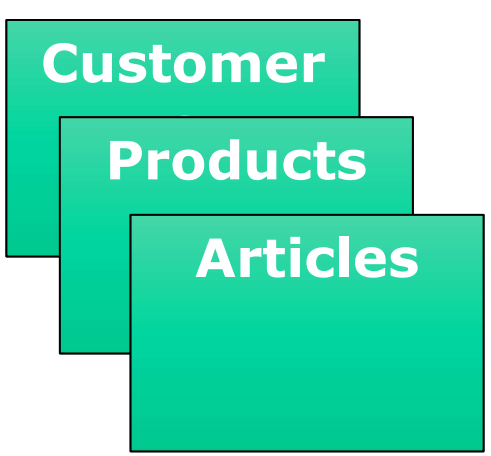
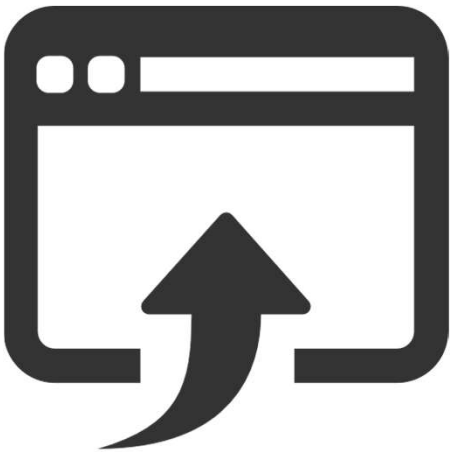
Wat is databinding

- Show – all kinds of – data in User Interface
- Data can come from:
  - Controller / class
  - Database
  - User input
  - Other systems





Declaratieve syntaxis

- Four (4) kinds of databinding
- Angular specific notation in HTML templates
    1. Simple data binding
    2. Event binding
    3. One-way data binding (Attribute binding)
    4. Two-way data binding

1. Simple Data binding

Class-properties binden in de template

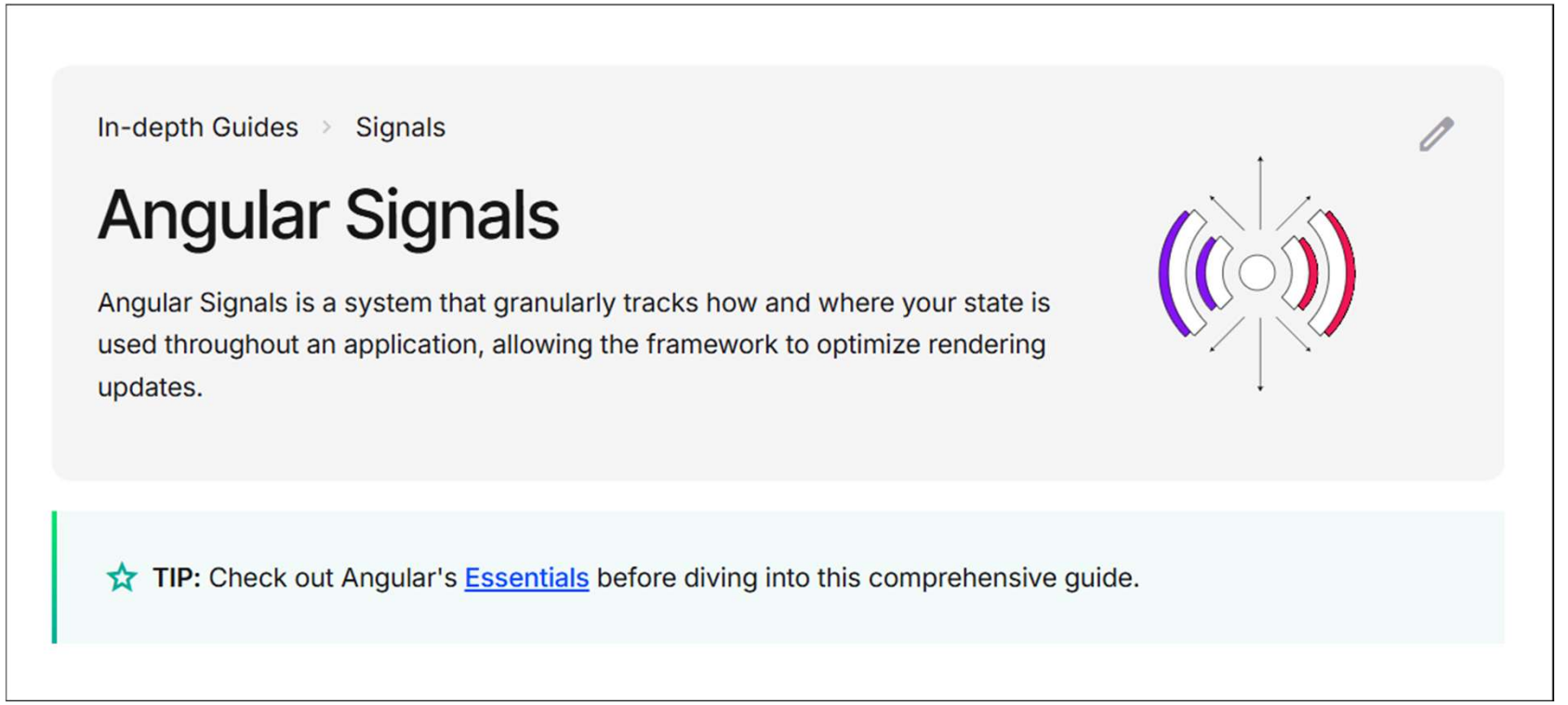
Variables: wrap them in a `signal()`

- Not mandatory – but best practice for reactivity and performance

*"A **signal** is a wrapper around a value that notifies consumers when that value changes. Signals can contain any value, from primitives to complex data structures."*

- You read a signal's value by calling its `getter` function, which allows Angular to track where the signal is used.
- Signals may be either *writable* or *read-only*.
- https://angular.dev/guide/signals

https://angular.dev/guide/signals



Creating and using signals

- In your TypeScript class:

```
// Signal holding the user's name as a string
protected name = signal<string>('Peter Kassenaar');

// Signal holding the current city as a string
protected city = signal<string>('Groningen');
```

1. Simple data binding syntax

Double curly braces and ‘invoking’ with signals:

```
<div>City: {{ city() }}</div>
```

```
<div>First name: {{ person().firstname }}</div>
```

Binding using a loop: `@for`

- Previously: `*ngFor="let item of items"`
- New: `@for (item of items; track item.id) {…}`
- ONLY: in Angular 17+ (👇 with classic variables)

```
cities : City[] = [
  {id: 1, name: 'Amsterdam', country: 'NL'},
  {id: 2, name: 'Berlin', country: 'GER'},
  {id: 3, name: 'Tokyo', country: 'JAP'},
]
```

```
<ul>
  @for (city of cities; track city.id) {
    <li>{{ city.id }} - {{ city.name }}</li>
  }
</ul>
```

- 1 - Amsterdam
- 2 - Berlin
- 3 - Tokyo

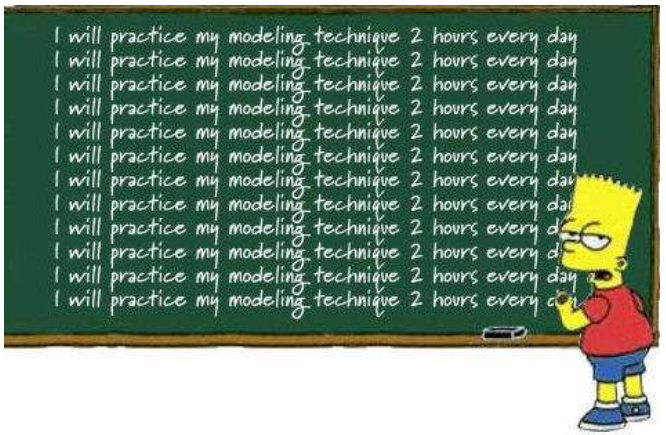
Or, using a signal to bind to a collection

```
// Signal containing an array of strings representing cities.
protected cities = signal<string[]>(['Groningen', 'Hengelo’,
                                     'Den Haag', 'Enschede']);
```

```
<h3>My favourite cities are:</h3>
<ul>
  @for (city of cities(); track $index) {
    <li>{{ city }}</li>
  }
</ul>
```

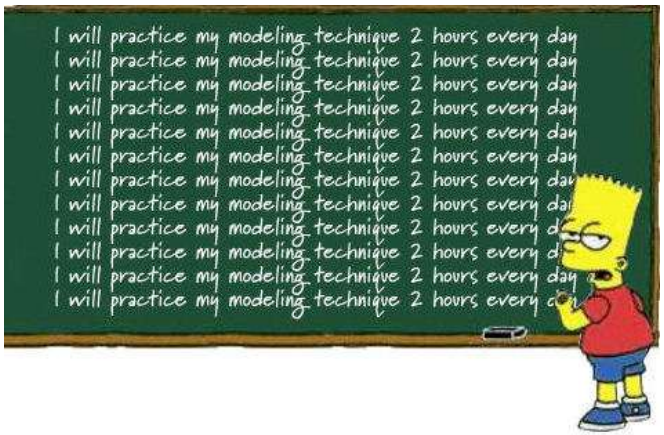
Workshop

- Expand your app from the previous lab (`Hello World`) with a field/property. Bind the property in the template being used.
    1. Use direct initialization of variables, like name: string = '<your-name>'.
    2. Use a signal for different types (`signal<string>, signal<number[]>,…`)
- The user interface will always be the same! The signal approach is just the best practice.
- Use additional properties.
- Demo `../101-databinding`



Workshop

- Create an array of properties. Bind them in the template using the directive `*ngFor` or `@for(…,;track …)`
  - What do you need to import for the first notation?
- Use TypeScript to explicitly declare the property as an array of strings.
  - `cities: WritableSignal<string[]>;` OR
  - `cities: WritableSignal<Array<string>>;`
- Of course you can use other data than cities, for example persons, products, and so on.
- Demo `../101-databinding`.



Checkpoint

- Simple data binding `{{ ... }}`
- Properties of the class are bound
- Best practice: wrap properties in signals
- You can bind them to the template
- Use an array of data to bind to the template
  - Use `*ngFor` or `@for()` to loop over data

Creating a Model (as in: MVC)

A Model as a class with exported public properties:

```
export class City{
   constructor(
      public id: number,
      public name: string,
      public province: string,
   ){ }
}
```

Notice shorthand notation `public id : number`

1. Defines a private/local parameter
2. Defines a public parameter with the same name
3. Initializes parameter at instantiation of the class with `new`

Using the Model

1. Import model class

```
import {City} from './city.model'
```

2. Update component

```
export class AppComponent {
      name   = 'Peter Kassenaar';
      cities =[
         new City(1, 'Groningen', 'Groningen'),
         new City(2, 'Hengelo', 'Overijssel'),
         new City(3, 'Den Haag', 'Zuid-Holland'),
         new City(4, 'Enschede', 'Overijssel'),
      ]
}
```

2a. Or: using signal:

```
protected cities = signal<City[]>([
  new City(1, 'Groningen', 'Groningen'),
  new City(2, 'Hengelo', 'Overijssel'),
  new City(3, 'Den Haag', 'Zuid-Holland'),
  new City(4, 'Enschede', 'Overijssel'),
]);
```

3. Update View

```
<ul>
  @for (city of cities(); track city.id) {
    <li>{{ city.id }} - {{ city.name }}</li>
  }
</ul>
```

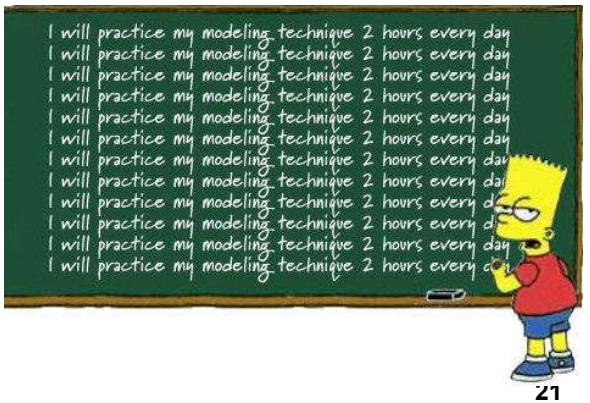
Another option: `interface`

- `Interface` only describes the structure of the data
- No keyword `new`
- No functionality in the 'instances'
- Mostly – personal preference!

```
interface ICity {
  id: number;
  name: string;
  province: string;
}
```

Workshop

- Create a Model for the contents of your array. The model consists of an object with one or more properties.
  - See for instance app/shared/city.model.ts as an example.
- Update the signature of the array so it looks like cities: `City[]` or cities: `Array<City>` in your variable or in your `signal<T>`
- Rewrite your array, so the content now are properties of type `<YourModel>` (often abbreviated with just `<T>`)
- Optional: instead of using a class for the Model you can also use the TypeScript-construct `interface` or `type`.
  - Look up for yourself how this can be done.



Binding conditionally using `@if`

- Previously: `*ngIf="…"`
- New: `@if (condition) { … }`
- ONLY : `Angular 17+`

Invalid

```
<div @if="showTitle">
  {{  title }}
</div>
```

Valid

```
@if (showTitle){
  <div>
    {{  title }}
  </div>
}
```

Classic `*ngIf` notation

Use the `*ngIf` directive (pay attention to the asterisk!)

```
<h2 *ngIf="cities.length > 3">There are a lot of favorite cities!</h2>
```

External templates

If you don’t like inline HTML :

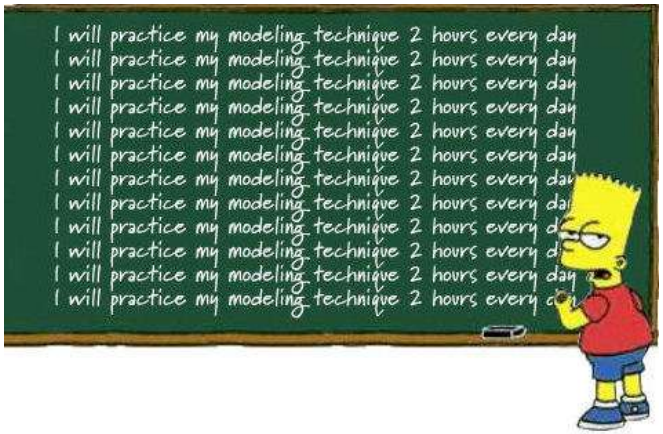
```
@Component({
   selector    : 'hello-world',
   templateUrl: 'app.component.html'
})
```

File `app.html`

```
<!-- HTML in external template -->
<h1>Hello Angular</h1>
<p>This is an external template</p>
<h2>My name is : {{ name }}</h2>
<h2>My favorite cities :</h2>
…
```

Workshop

- Create a `<div>` on the page that is only shown if your array has three or more objects in it.
  - The code can look like `<div *ngIf="cities.length > 3">…</div>`.
- Create one more use case of `*ngIf` for yourself.
  - Next: use `@if()` modern syntax.
- Move the expression to the TypeScript-part of the application (where it belongs!). Let it evaluate to a true | false value and bind this value via a property in the UI.



User input and event binding

React to mouse, keyboard,

hyperlinks and more

Event binding syntax

Angular: use parentheses for capturing and handling events:

Angular 1.x:

```
<div ng-click="handleClick()">…</div>
```

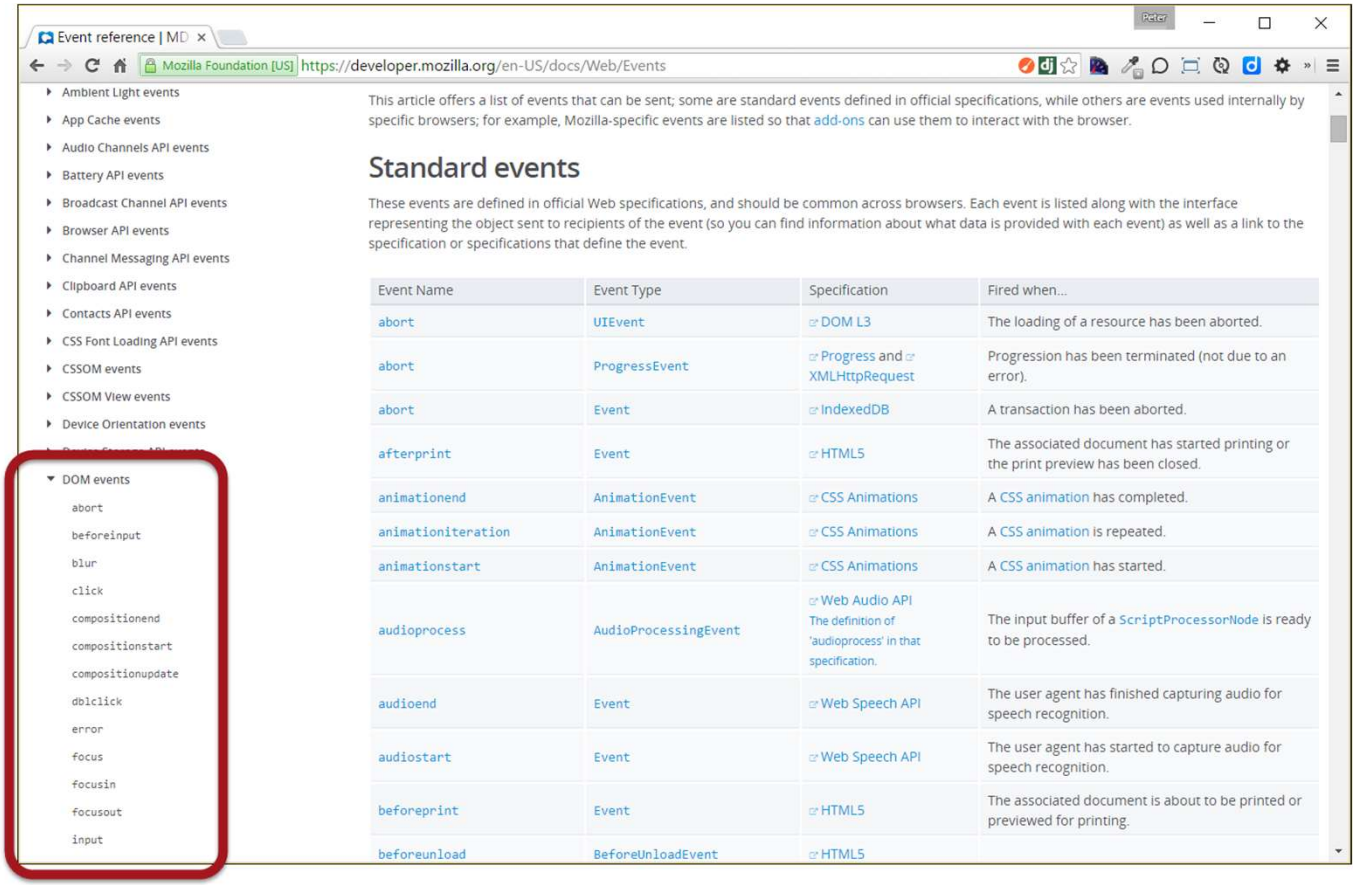
Angular:

```
<div (click)="handleClick()">…</div>
```

```
<input (blur)="onBlur()" />
```

DOM-events

- Angular can listen to *any* DOM-event without needing different directives:



https://developer.mozilla.org/en-US/docs/Web/Events

Example event binding

HTML

```
<!-- 1. Event binding for a button -->
<button class="btn btn-success"
        (click)="btnClick()">I am a button
</button>
<!--    Current value of the counter signal is written in the UI:-->
<p>You clicked: {{ counter() }} time(s).</p>
<hr>
```

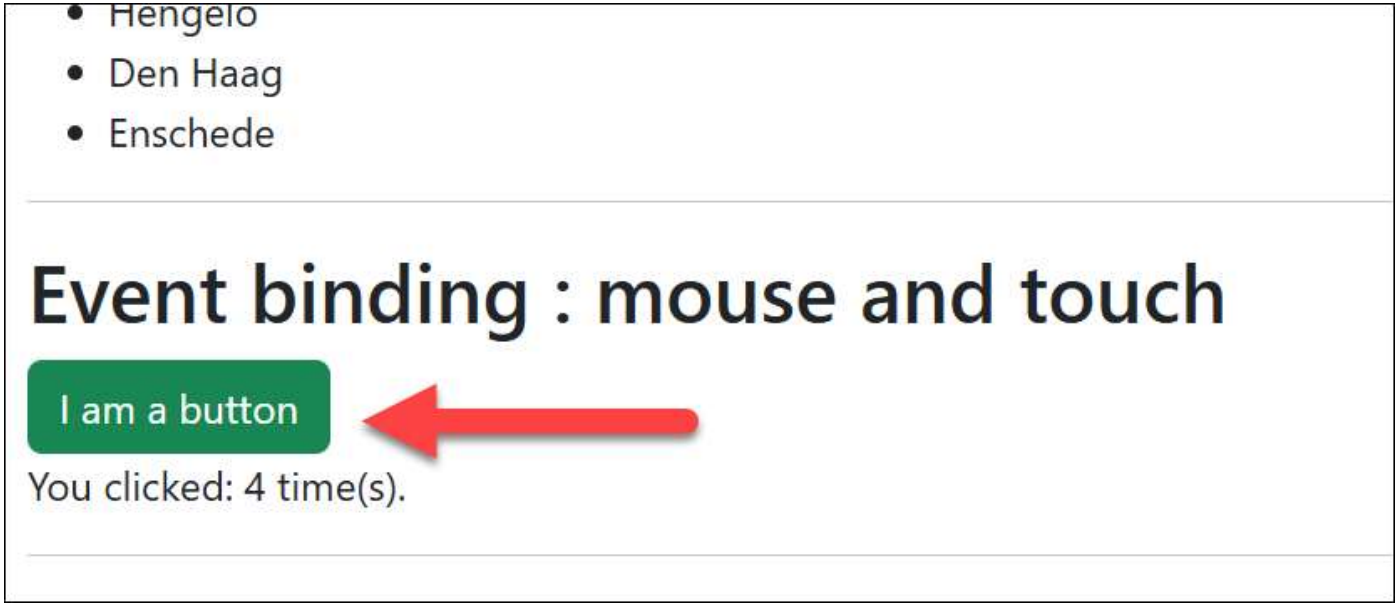
Class

```
protected counter = signal<number>(0);

// 1. Bind to click-event in the page
btnClick() {
  this.counter.update(value => value + 1);
}
```

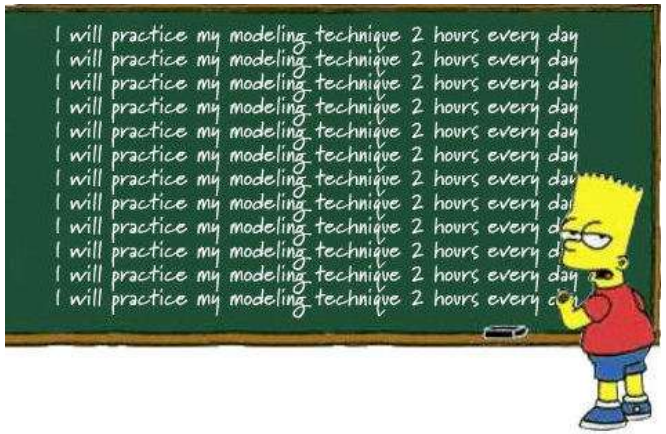
Remember to call `counter.update()`. It is invalid to update counter directly like `this.counter++`

Result


Hengelo
Den Haag
Enschede
Event binding : mouse and touch
I am a button
You clicked: 4 time(s).

Workshop

- Add an element with event-binding to the application. For instance, create a button to capture a `click` event.
- Call an event handler in the component if the event occurs.
  - For example, write a function `handleClick() { alert('message…')}` or show a `console.log()`-text.
- Next: show the the message via a variable in the UI. It should read something like '`you clicked on <your-entity-name>`'.
- Demo .../`103-eventbinding`..



Checkpoint

- Event binding is done with `(eventname)="…"`
- Events are always notated in lowercase.
- You can bind multiple events to the same element.
- Events are *not* rendered in the browser DOM-tree
- Events are handled by an event handler-function on the component

Reading values from text fields

Creating a variable from your text field

A) Event parameters: $event

HTML

```
<input type="text" class="input-lg" placeholder="City..."
       (keyup.enter)="onKeyUp($event)"><br>
<p>{{ txtKeyUp}}</p>
```

```
// 2. Bind to keyUp-event in the textbox
onKeyUp(event:any){
   this.txtKeyUp = event.target.value + ' - ';
}
```

B) Event parameters local template variable

Declare *local template variable* with # → The complete element is passed to the component

```
<input type="text" class="input-lg" placeholder=”City..."
      #txtCity (keyup)="betterKeyUp(txtCity)">
<h3>{{ txtCity.value }}</h3>
```

Class:

```
 // 3. Bind to keyUp-event via local template variable
betterKeyUp(txtCity){
   //... Handle txtCity as desired
}
```

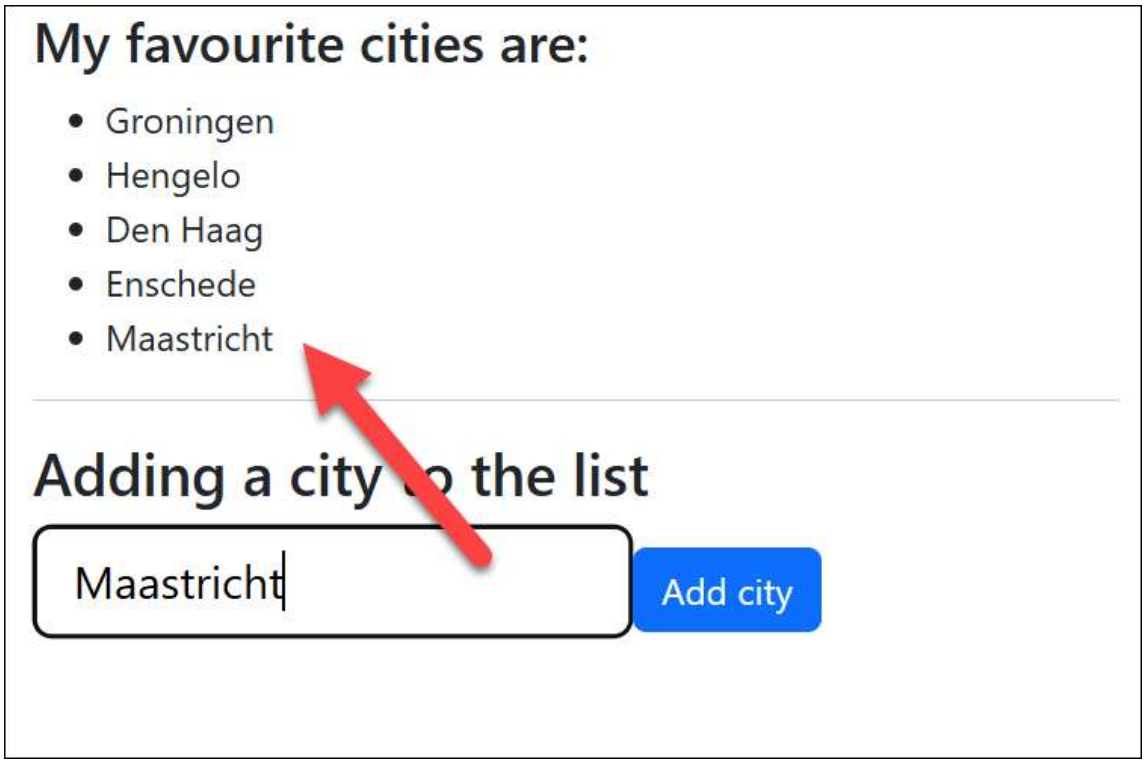
Putting it all together…

HTML

```
<input type="text" class="input-lg" placeholder="City..." #txtCity>
<button class="btn btn-success"
      (click)="addCity(txtCity)">Add city
</button>
```

Class

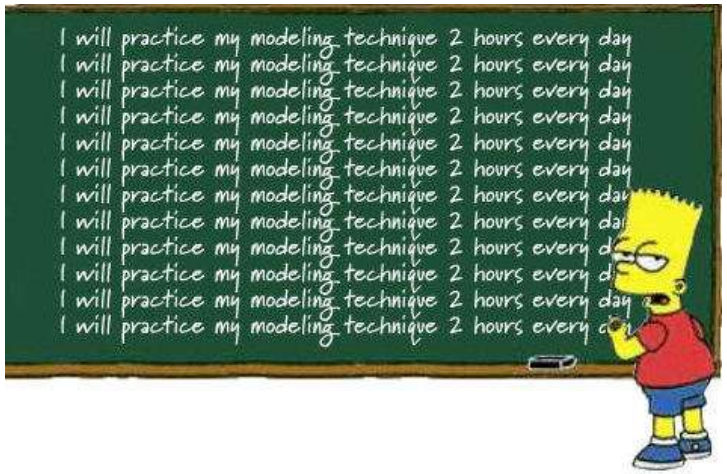
```
export class AppComponent {
   // Properties on component/class
   …
   addCity(txtCity) {
      let newID   = this.cities.length + 1;
      let newCity = new City(newID, txtCity.value, 'Unknown');
      this.cities.update(newCity);
      txtCity.value = '';
   }
}
```



We call *the same* method `addCity()` from `(keyup.enter)` for the textbox, as the `(click)` event for the button.

Workshop

- Create a text field with a local template variable.
  - Pass the variable to an event handler using an event of your liking (for example click or keyup) and show the value in an `alert()` or `console.log()`
- Create another textbox on the page. The user can type numbers in the textbox.
  - Pass the number to an event handler and add the number to a property `total`.
  - Show the addition of all numbers (i.e. the total value) in the page.
- Remember to use `parseInt()` to convert the stringvalue of the textbox to a number!
- Demo `…/103-eventbinding`.



Checkpoint

- Event binding is addressed with `(eventname)="…"`
- Events are being handled by a function inside the component
- Optional: use `$event` to pass data to the class
- Or: use a local template variable `#` to pass value to the class
- You can create simple, client sided CRUD-operations this way.

Attribute & property binding

Bind values dynamically to HTML attributes and DOM-properties

Attribute binding syntax

- Bind directly to properties of HTML-elements.
- Also know as *one-way binding.*
- Use square brackets syntax

Angular 1.x:

```
<div ng-hide="true|false">…</div>
```

Angular:

```
<div [hidden]="true">…</div>
```

Or :

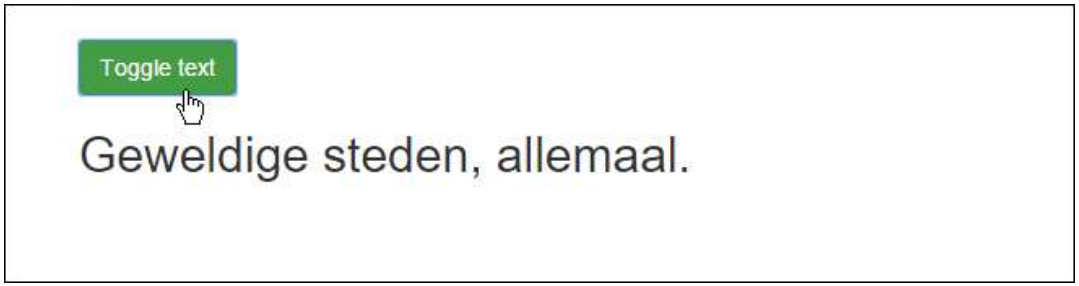
```
<div [hidden]="person.hasEmail">…</div>
<div [style.background-color]="'yellow'">…</div>
```

Example attribute binding

HTML

```
<!-- Attribute binding -->
<button class="btn btn-success" (click)="toggleText()">Toggle text</button>
<h2 [hidden]="textVisible">I love all these cities!</h2>
```

```
// Toggle attribute: show or hide text.
toggleText(){
   this.textVisible = !this.textVisible;
}
```



For instance...

HTML

```
@for (city of cities(); track city.id) {
  <li class="list-group-item"
      (click)="updateCity(city)">
    {{ city.id }} - {{ city.name }}
  </li>
}
```

Class

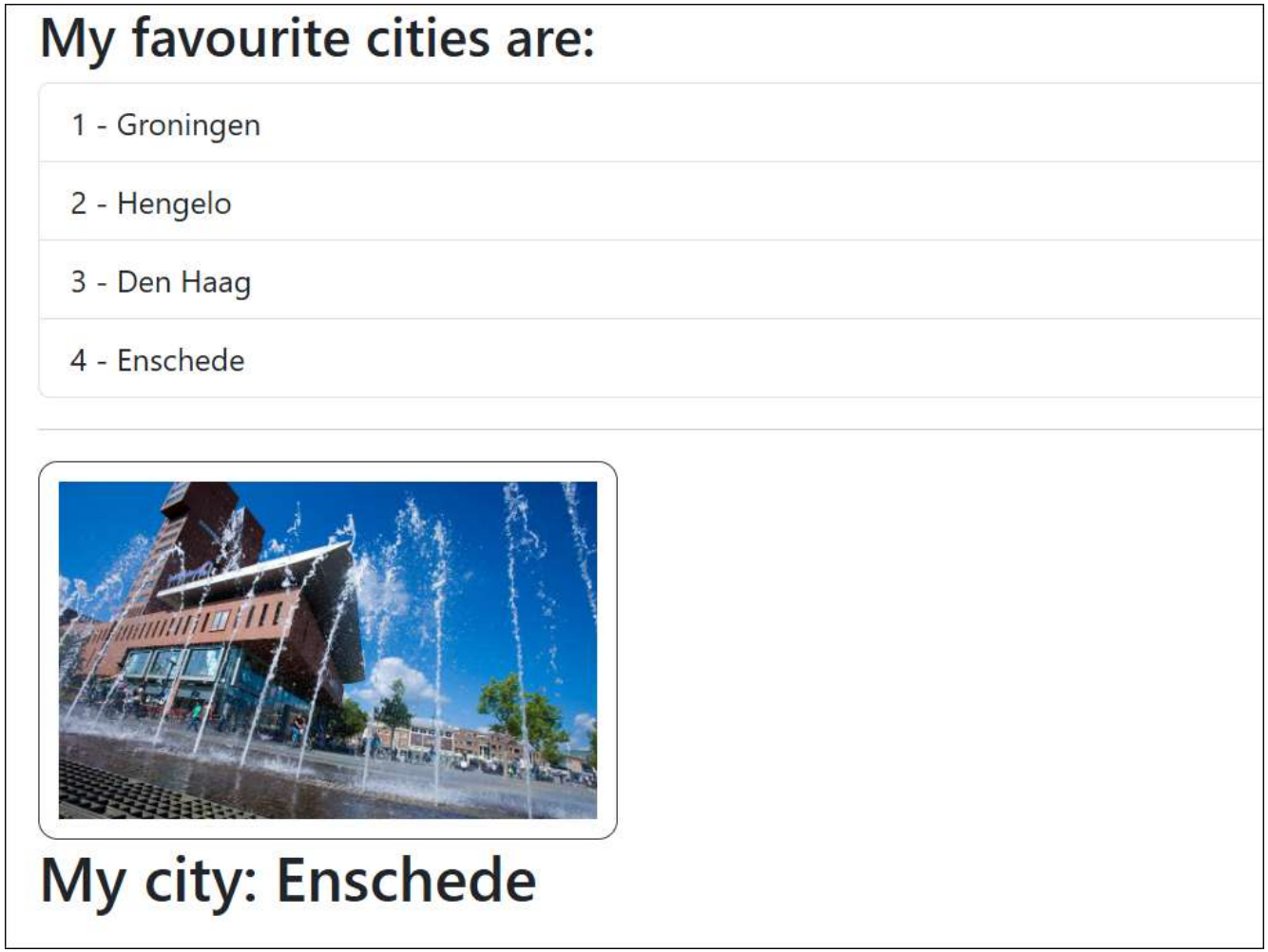
```
protected currentCity = signal<City|null>(null);
protected cityPhoto = signal<string>('');

updateCity(city: City) {
  console.log(`Updating city: ${city.name}`);
  this.currentCity.set(city);

  const imageFilename = `assets/img/${city.name}.jpg`;
  this.cityPhoto.set(imageFilename);
}
```

Demo:

`..\103-attributebinding\src\app\app.ts`



More information : https://angular.dev/guide/templates/binding#css-class-and-style-property-bindings

Workshop

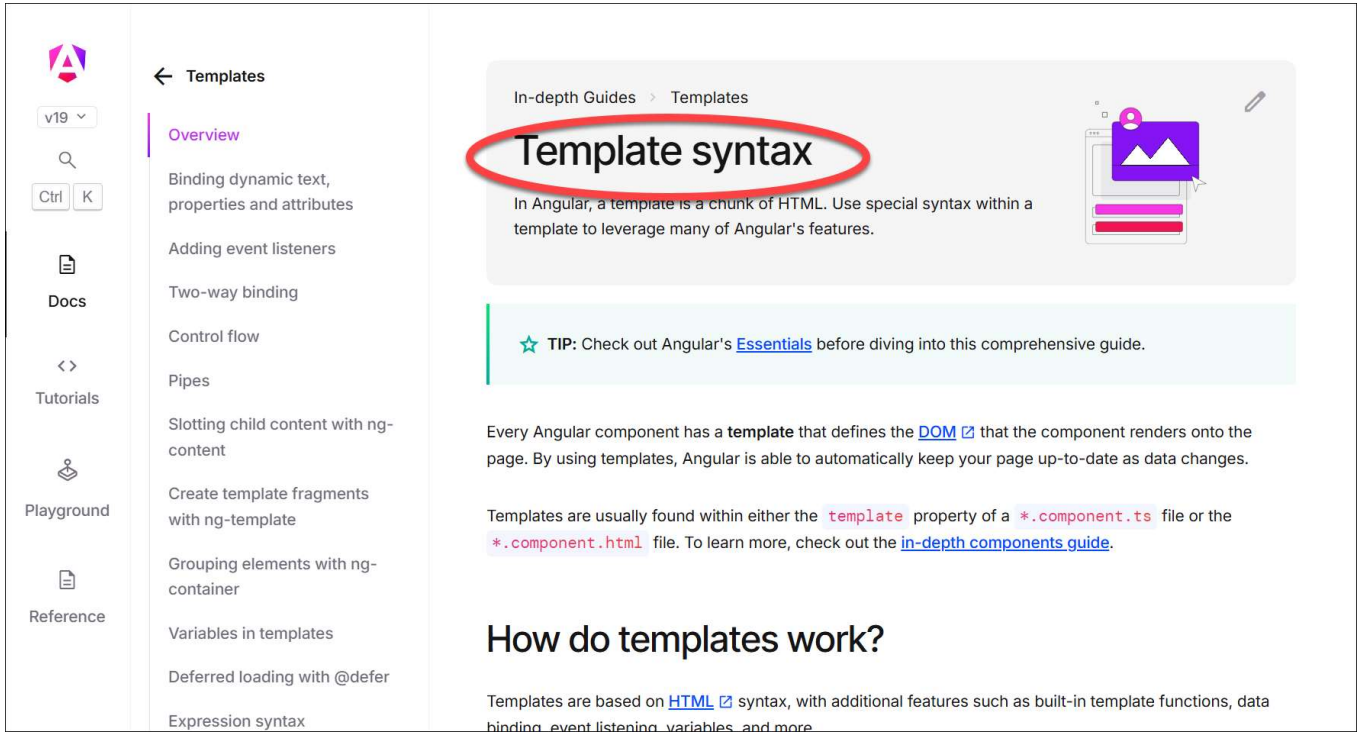
- Create a button on the page. If the button is clicked, a `<div>` with a text is shown.
- If the button is clicked again, the text is hidden.
  - Demo code available at `…/103-attributebinding`
- Optional: create a component with a textbox.
  - If the user types an English color in the box and clicks a button or presses `Enter`, a corresponding receives this color as a background color.
  - Hint: use `[style.backgroundColor]="bgCcolor"` on the `div`. Create a `bgCcolor` property on the class.
- Optional: create a second textbox to set the text/foreground color.
- Advanced: investigate how this works if the color can be picked from a series of radio buttons or from a dropdown list.

Checkpoint

- Attribute binding is addressed with `[attrName]="…"`
- Attributes are bound to a variable on the class.
- You can calculate the variable in the `.ts`-file

More binding-options

- Attribute binding and DOM-property binding: `[…]`
- Class binding : `[ngClass](now deprecated)`
- Style binding : `[ngStyle](now deprecated!)`
- https://angular.dev/guide/templates



Two-way binding

Updating user interface and class variables at the same time

Two way binding syntaxis

Was removed from Angular for a while, but returned after complaints from the community:

Angular 1.x:

```
<input ng-model="person.firstName" />
```

Angular: similar, but notation is a little bizar:

```
<input [(ngModel)]="person.firstName" />
```

Using [(ngModel)]

```
<input type="text" class="input-lg" [(ngModel)]="newCity" />
<h2>{{ newCity }}</h2>
```

Which is shorthand-notation for:

```
<!-- Two-way binding with extended syntax -->
<input type="text" class="input-lg"
       [value]="newCityExtended"
       (input)="newCityExtended = $event.target.value" />
<h2>{{ newCityExtended }}</h2>
```

FormsModule importeren

- Two-way binding used to be in the Angular Core – now in it's own module
  - Import `FormsModule` in your component
  - Classic: import in `app.module.ts`

- `import {FormsModule} from "@angular/forms";`
- …
- `imports      : [FormsModule, …],`

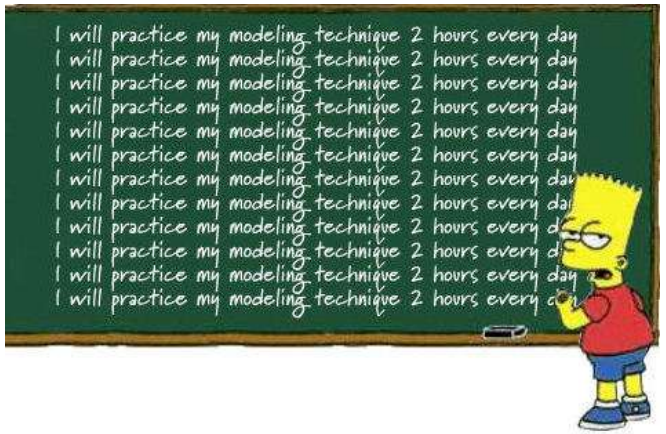
Our Goal: passing data from View to Controller,

We have lots of options:

1. Using `$event`
2. Using a Local Template Variabele `#NameVar`
3. Using `[(ngModel)]` (to be used in simple situations, mostly not on complex forms)
4. HostBinding/`@HostListener` (via @-decorators)
5. Use `@ViewChild()` …

Workshop

- Create a text field in your component that uses two-way binding.
  - Use `[(ngModel)]` as a directive on the `<input>` box. Bind the value of the typed text directly to the page.
- Create a copy-function. Create two text boxes on the page.
  - Text that is typed into the first textbox, should also appear in the second textbox.
- Demo …/`104-twowaybinding`



Checkpoint

- Two-way binding is addressed with **`[(ngModel)]="…"`**
- The value of `[(ngModel)]` is updated *automagically* by Angular.
- Use `[(ngModel)]` on `<input type="..." />` tags
- The directive is available in the View/Template and in the TypeScript class.
  - Changes you make in the template are reflected in your class
  - Changes you make in your class are reflected in the template

Declarative syntax

- Four (4) types of databinding
- Angular specific notation in HTML templates
  1. Simple data binding with {{ … }}
  2. Event binding with (…)
  3. One-way data binding (Attribute binding) with […]
  4. Two-way data binding with [(ngModel)]="…"

Checkpoint

- Databinding in Angular is different from other frameworks
- Learn the syntax on DOM- and Attribute binding. Also learn event binding en two-way binding.
- Optional: host binding with `@HostListener()`
- Always edit the class and corresponding View
- A lot of concepts are the same
  - However, the way to achieve results are different in Angular, compared to Vue, React and other frameworks.